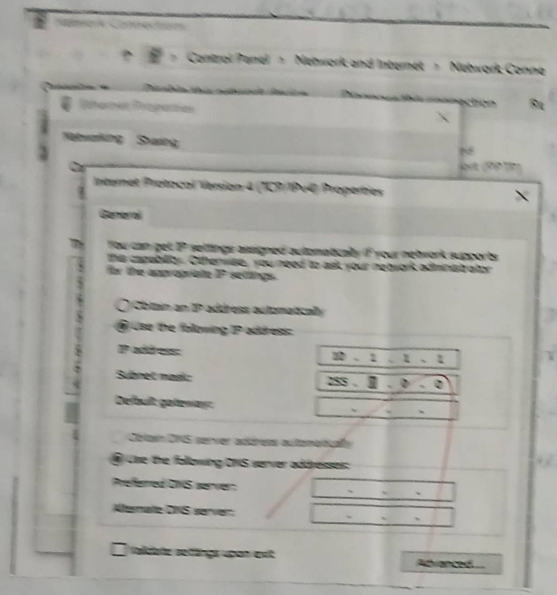


EX.No:8 SETUP AND CONFIGURE A LOCAL AREA NETWORK USING A SWITCH AND ETHERNET CABLE IN YOUR LAB

Aim:

To setup and configure a Lan using a switch and ethernet cable in your lab.

What is Lan:

A Lan is a network that connects devices within a small area, such as a home, school or office. It allows devices like computers, printers and servers to share files, printers and internet access.

How to setup a Lan:

1) Plan and design an appropriate network topology taking into account network requirements and equipment location.

2) you can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes and 4 ethernet cables.

3) connect your computers to network switch via an ethernet cable, which is a simple a plugging one end of the ethernet cable into your computer and the other end into your network switch.

4) Assign IP address of your Pc's.

1. Log on to the client computer as Admin or as owner
2. click Network and Internet connection
3. Right click Local area connection / Ethernet → click to properties → select Internet protocol → click on properties → select use the following ip address option and assign ip address.
4. click the properties → select use the following ip address on option and assign ip address.

5) configure a network switch.

1. Connect your computer to the switch. To access the switch's web interface, you will need to connect your computer to the switch using a Ethernet cable.
2. Log in to web interface: open a web browser and enter IP address of the switch in the address bar.
3. configure basic settings: once you're logged in, you will be able to configure basic settings for the switch
4. Assign IP address as: 10 : 111.5 . subnet make 255.0.0.0.

6) check the connectivity between switch and other machine by using ping command in the command prompt of the device.

7) select a folder → go to properties → click sharing tab → share it with everyone the same Lan.

8) Try to access the shared folder from others computers of the network

(90)

Result:

Thus, the setup and configure a LAN using a switch and ethernet cables in your lab was completed successfully.